

## 5 Time Series Response

### 5.1 Transfer Functions Block Diagrams

Transfer functions can be created using:

- Polynomial model (numerator / denominator)

$$G(s) = \frac{B(s)}{A(s)} = \frac{b_ms^m + b_{m-1}s^{m-1} + \dots + b_0}{a_ns^n + a_{n-1}s^{n-1} + \dots + a_0} \quad (1)$$

- $n$  = order of transfer function models where  $m < n$

- Zero-pole-gain model (numerator / denominator)

$$G(s) = k \frac{(s - z_1)(s - z_2) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_n)} \quad (2)$$

- zeros:  $z_1, z_2, \dots, z_m$  roots of  $B(s) = 0$
- poles:  $p_1, p_2, \dots, p_n$  roots of  $A(s) = 0$
- gain:  $k$

- Time constant model

$$G(s) = \frac{k}{s^N} \cdot \frac{(T_a + 1)(T_b + 1) \cdots}{(T_1 + 1)(T_2 + 1) \cdots} \quad (3)$$

- $N$  = type of transfer function model

#### 5.1.1 Create transfer function using numerator and denominator coefficients

MATLAB can be used to create continuous-time single-input, single-output (SISO) transfer functions from their numerator and denominator coefficients using `tf`. To use the `tf` function, you must have the Control System Toolbox licensed and installed. To find out if you do, type: `ver control` in your Command Window or a script.

##### Method I

Create the transfer function

$$G(s) = \frac{s}{s^2 + 3s + 2} \quad (4)$$

Listing 1: MATLAB code for Method I.

```
1 num = [1 0];  
2 dem = [1 3 2];  
3 G = tf(num, dem)  
4 % To use the tf function, you must have the Control System Toolbox  
5 % licensed and installed. To find out if you do, type:  
6 % ver control in your Command Window or a script.
```

where `num` and `dem` are the numerator and denominator polynomial coefficients in descending powers of  $s$ . For example, `dem = [1 3 2]` represents the denominator polynomial  $\frac{s}{s^2 + 3s + 2}$

`G` is a `tf` model object, which is a data container for representing the transfer function in polynomial form.

##### Method II

Alternatively, you can specify the transfer function  $G(s)$  as an expression in  $s$ -domain.

1. Create a transfer Function model for the variable  $s$
2. Specify  $G(s)$  as a ratio of polynomials in  $s$

Listing 2: MATLAB code for Method II.

---

```

1 s = tf('s');
2 G = s/(s^2+3*s+2)

```

---

Therefore, the full expression of  $G(s)$  can be written as

$$G(s) = \frac{B(s)}{A(s)} = \frac{s}{s^2 + 3s + 2} = \frac{b_1s + b_0}{a_1s^2 + a_2s + a_0} \quad (5)$$

where

$$B(s) = b_1s + b_0 \quad (6)$$

resulting in  $b_1 = 1$ , and  $b_0 = 0$ ; or,  $B = [1 \ 0]$ . Moreover,

$$A(s) = a_1s^2 + a_2s + a_0 \quad (7)$$

where  $a_1 = 1$ ,  $a_2 = 3$ , and  $a_0 = 1$ ; or,  $A = [1 \ 3 \ 1]$ .

### 5.1.2 Create transfer function using Zeros, Poles, and Gain

MATLAB can be used to create continuous-time single-input, single-output (SISO) transfer functions in factored form using `zpk`. Create the factored transfer function

$$G(s) = 5 \frac{s}{(s-1-i)(s-1-i)(s-2)} \quad (8)$$

Listing 3: MATLAB code to create a transfer function using Zeros, Poles, and Gain.

---

```

1 Z = 0;
2 P = [-1-1i -1+1i -2];
3 K = 5;
4 G = zpk(Z,P,K)

```

---

where  $Z$  and  $P$  are zeros and poles (the roots of the numerator and denominator respectively).  $K$  is the gain of the factored form. Solving for the poles  $p_1$ ,  $p_2$ , and  $p_3$  of  $G(s)$ ;

$$G(s) = 5 \frac{s}{(s-1-i)(s-1-i)(s-2)} \quad (9)$$

$$= 5 \frac{s-0}{[s-(-1-i)][s-(-1-i)][s-(-2)]} \quad (10)$$

where  $K = 5$ ,  $s-0 = s-z_1 \rightarrow z_1 = 0$ , and  $Z = [0]$ . Therefore,

$$[s-(-1-i)][s-(-1-i)][s-(-2)] = (s-p_1)(s-p_2)(s-p_3) \quad (11)$$

this leads to

$$G(s) = k \frac{s-z_1}{(s-p_1)(s-p_2)(s-p_3)} \quad (12)$$

therefore,  $G(s)$  has a real pole at  $s = -2$  and a pair of complex poles as  $s = -1 \pm i$ . The vector  $P = [-1-1i \ -1+1i \ -2]$  specifies these pole locations.

## 5.2 Order verse Type

A system has a “Type” and an “Order”, which have different meanings.

- Order =  $n$ , highest exponent of  $s$  in the denominator.  $n$  is the number of poles.
- Type =  $N$ , exponent of the factored out  $s$  in the denominator.  $N$  is the number of poles in origin ( $p = 0$ ).

Consider the 1<sup>st</sup>-order mass-damper system (no stiffness) as shown in figure X with the transfer function

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s} \quad (13)$$

The transfer function can easily be written in the basic form

$$G(s) = \frac{b_0}{a_2 s^2 + a_1 s} \quad (14)$$

where  $a_2 = 1$ ,  $a_1 = 2\zeta\omega_n$ ,  $b_0 = \omega_n^2$ . There the presence of  $a_2$  means its a 2<sup>nd</sup> order system. To solve for the type of the system,  $s$  must be factored of of the denominator, leading to:

$$\begin{aligned} G(s) &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s} \\ &= \frac{\omega_n^2}{s(s + 2\zeta\omega_n)} \\ &= \frac{\frac{\omega_n}{2\zeta}}{s} \cdot \frac{1}{\frac{1}{2\zeta\omega_n}s + 1} \\ &= \frac{K}{s^N} \cdot \frac{1}{T_1 s + 1} \end{aligned} \quad (15)$$

where  $K = \frac{\omega_n}{2\zeta}$ ,  $N = 1$ , and  $T_1 = \frac{1}{2\zeta\omega_n}$ .  $N = 1$  means that is a Type 1 system. Therefore, this is a 2<sup>nd</sup> order system of Type 1, “Type” and “Order” have different meanings. Table ?? reports the types and orders for different transfer functions.

need to find some real-world examples of orders and types.

make a figure

Table 1: Examples of types and orders for different transfer functions.

| transfer function                                                                                                                                                                                       | Type | Order |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| $G(s) = \frac{1}{Ts+1}$                                                                                                                                                                                 | 0    | 1     |
| $G(s) = \frac{1}{cs} = \frac{1/c}{s}$                                                                                                                                                                   | 1    | 1     |
| $G(s) = \frac{1}{Js^2} = \frac{1/J}{s^2}$                                                                                                                                                               | 2    | 2     |
| $G(s) = \frac{1}{Js^2+cs} = \frac{K/c}{s} \cdot \frac{1}{J/c s+1}$                                                                                                                                      | 1    | 2     |
| $G(s) = \frac{K}{J^4} \frac{T_a s + 1}{(T_1 s + 1)(T_2 s + 1)(T_3 s + 1)}$ $= \frac{b_1 s + b_0}{s^4(a_3 s^3 + a_2 s^2 + a_1 s + a_0)}$ $= \frac{b_1 s + b_0}{(a_3 s^7 + a_2 s^6 + a_1 s^5 + a_0 s^4)}$ | 4    | 7     |

### 5.3 Time Response

Time response calculations are obtained using the Laplace transforms where the Laplace transform is

$$X(s) = G(s)F(s) \quad (16)$$

and the time response is

$$x(t) = \mathcal{L}[X(s)]^{-1} \quad (17)$$

#### 5.3.1 Time Series Response for a Step Function

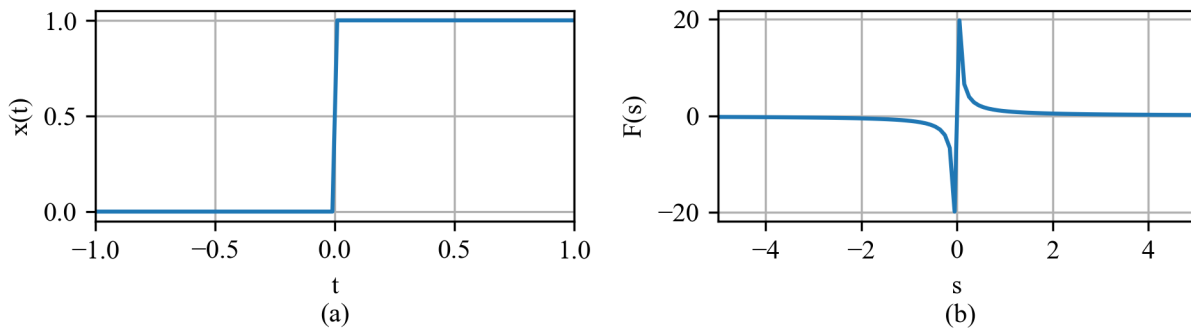


Figure 5.1: Step function; showing the (a) time domain and; (c) the s-space.

A step function is expressed as the following Laplace pair:

$$\text{LT pair} = \begin{cases} f(t) & 1(t), \quad t > 0 \\ F(s) & \frac{1}{s} \end{cases} \quad (18)$$

therefore, the time response of the systems is expressed as

$$x(t) = \mathcal{L} \left[ G(s) \frac{1}{s} \right]^{-1} \quad (19)$$

In MATLAB, this is expressed as:

Listing 4: MATLAB code for the time-series response of a step function.

---

```
1 x_t = step(G)
```

---

### 5.3.2 Time Series Response for a Impulse Function

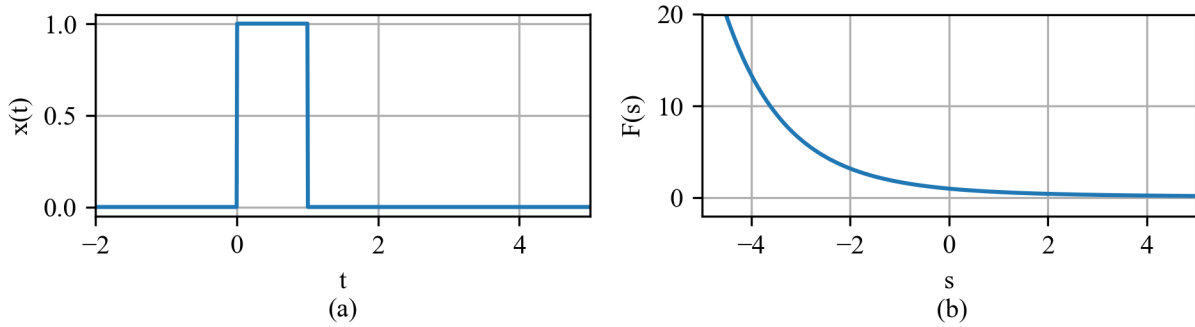


Figure 5.2: Pulse function; showing the (a) time domain and; (c) the s-space.

An impulse function is expressed as the following Laplace pair:

$$\text{LT pair} = \begin{cases} f(t) & p(t; \tau) \\ F(s) & \frac{1 - e^{-st\tau}}{s\tau} \end{cases} \quad (20)$$

therefore, the time response of the systems is expressed as

$$x(t) = \mathcal{L} \left[ G(s) \frac{1 - e^{-st\tau}}{s\tau} \right]^{-1} \quad (21)$$

In MATLAB, this is expressed as:

Listing 5: MATLAB code for the time-series response of a step function.

---

```
1 x_t = impulse(G)
```

---

### 5.3.3 Time Series Response for a Ramp Function

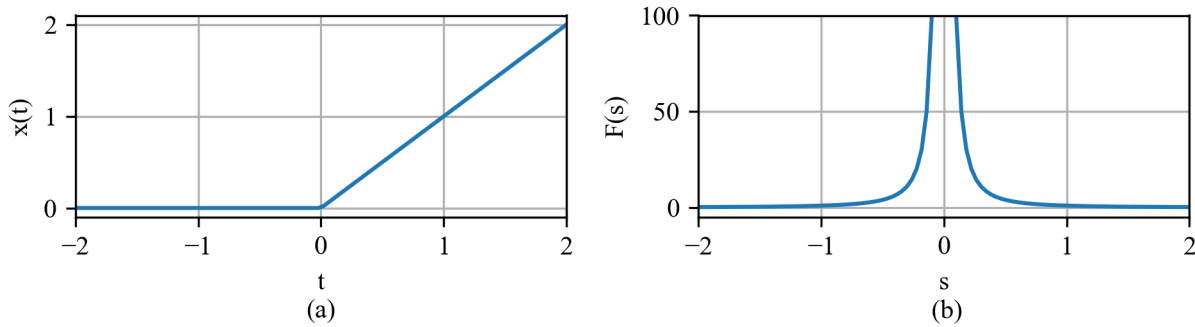


Figure 5.3: Ramp function; showing the (a) time domain and; (c) the s-space.

An ramp function is expressed as the following Laplace pair:

$$\text{LT pair} = \begin{cases} f(t) & t, \quad t > 0 \\ F(s) & \frac{1}{s^2} \end{cases} \quad (22)$$

therefore, the time response of the systems is expressed as

$$x(t) = \mathcal{L} \left[ G(s) \frac{1}{s^2} \right]^{-1} \quad (23)$$

In MATLAB, this is expressed as:

Listing 6: MATLAB code for the time-series response of a step function.

---

```
1 x_t = impulse(G/(s^2))
```

---

Note that for the MATLAB code, we used the property:

$$X(s) = G(s) \frac{1}{s^2} = \left( \frac{G(s)}{s^2} \right) \cdot 1 \quad (24)$$

where 1 is the Laplace transform of an impulse. Note that ramp is not an option in MATLAB as this command is already used to generate a time-series ramp signal.

## 5.4 1<sup>st</sup> Order System Time Response

The first order equation of motion is

$$T\dot{x}(t) + x(t) = f(t) \quad (25)$$

where  $x(0) = 0$  is the initial condition and  $T$  is a time constant for the first order system. The Laplace transform gives us

$$\begin{aligned} x &\rightarrow X(s) \\ \dot{x} &\rightarrow sX(s) \\ f(t) &\rightarrow F(s) \end{aligned} \quad (26)$$

therefore, the s-domain equation is:

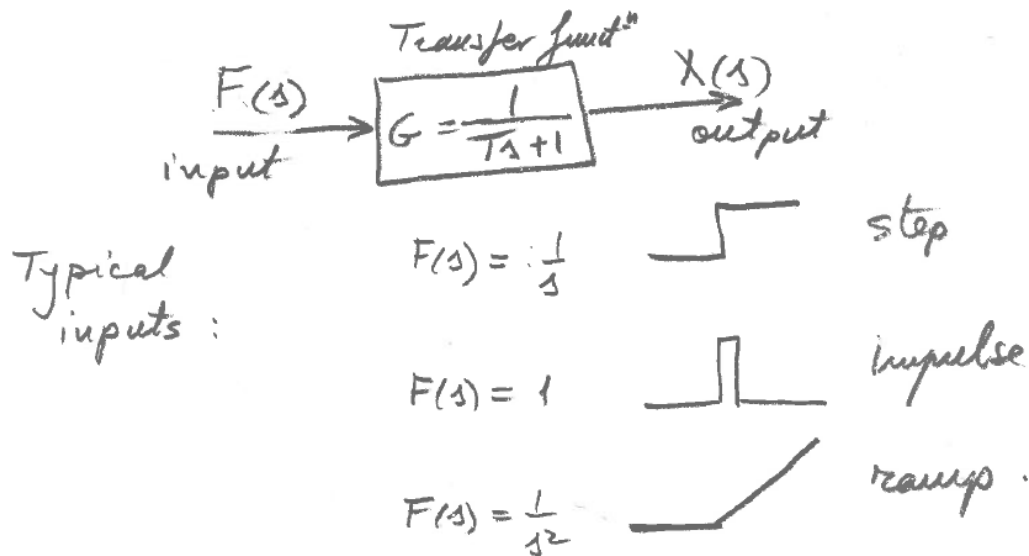
$$TsX(s) + X(s) = F(s) \quad (27)$$

where:

$$X(s) = \frac{F(s)}{Ts+1} = \frac{1}{Ts+1} F(s) = G(s)F(s) \quad (28)$$

therefore, the transfer function is:

$$G(s) = \frac{1}{Ts+1} \quad (29)$$



#### 5.4.1 Step response of a 1<sup>st</sup> Order System

$$X(s) = G(s)F(s) \quad (30)$$

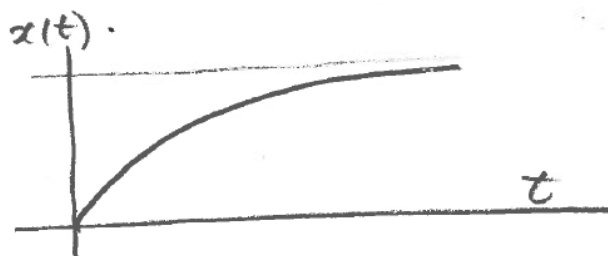
$$= \frac{1}{Ts+1} \cdot \frac{1}{s}$$

$$= \frac{1}{s(Ts+1)} \quad (31)$$

Therefore, solving for  $\mathcal{L}[X(s)]^{-1}$  yields

$$x(t) = 1 - e^{-t/T} \quad (32)$$

or



### 5.4.2 Impulse response of a 1<sup>st</sup> Order System

$$X(s) = G(s)F(s) \quad (33)$$

$$= \frac{1}{Ts+1} \cdot 1$$

$$= \frac{1}{Ts+1} \quad (34)$$

Therefore, solving for  $\mathcal{L}[X(s)]^{-1}$  yields

$$x(t) = \frac{1}{T}e^{-t/T} \quad (35)$$

or



### 5.4.3 Ramp response of a 1<sup>st</sup> Order System

$$X(s) = G(s)F(s) \quad (36)$$

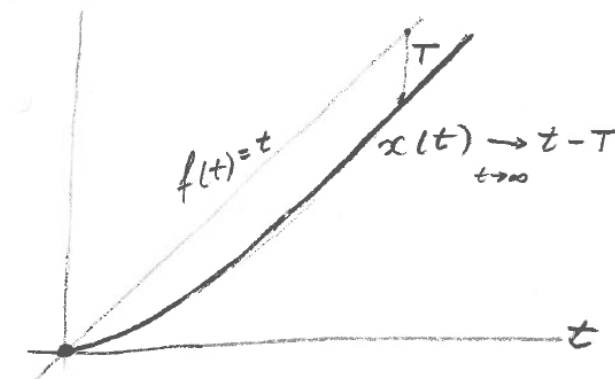
$$= \frac{1}{Ts+1} \cdot \frac{1}{s^2}$$

$$= \frac{1}{s^2(Ts+1)} \quad (37)$$

Therefore, solving for  $\mathcal{L}[X(s)]^{-1}$  yields

$$\begin{aligned} x(t) &= t - T + Te^{-t/T} \\ &= t - T(1 - e^{-t/T}) \end{aligned} \quad (38)$$

or



Moreover,

$$x(t) = t - T + Te^{-t/T} \xrightarrow{t \rightarrow \infty} t - T \quad (39)$$

#### 5.4.4 Summary of the First Order System Responses

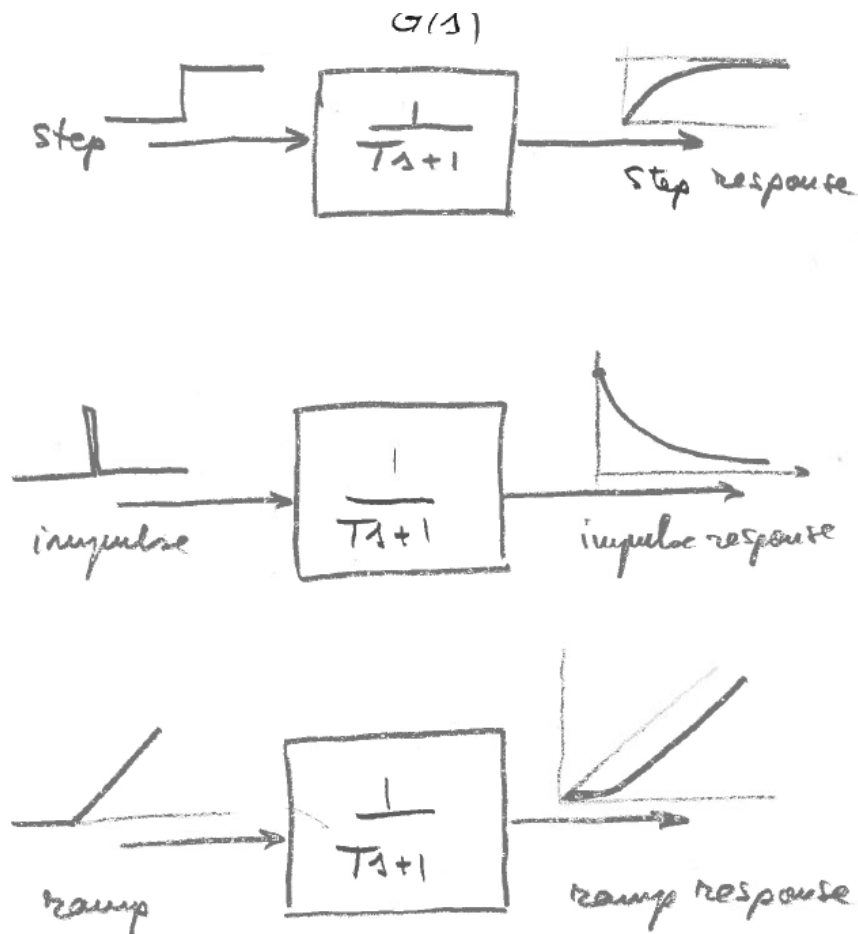


Figure 5.4: A summary of the first order system responses.

Listing 7: MATLAB code for time series responses of 1<sup>st</sup> order system.

```
1  %{
2  This program studies time response of 1st order systems
3  %}
4  clc
5  clear
6  close all
7
8  format compact
9
10 %% Given data
11 T=2.5; % time response for the 1D system
12
13 %% time range setup
14 T_max = 10; % run the test to 10 seconds
15 dt = T_max*1e-4; % find the delta-t value
16 t = 0:dt:T_max; % build the time vector
17
18 %% Define system
19 B = [1];
20 A = [T 1];
21 G = tf(B,A);
22
23 %% create the figure environment
24 figure(1)
25
26 %% step response
27 subplot(3,1,1)
28 hold on
29 step(G,t)
30 ylim([0 1.2])
31
32 %% Impulse response
33 subplot(3,1,2)
34 impulse(G,t)
35
36 %% Ramp response
37 F_ramp = tf([1],[1 0 0])
38 subplot(3,1,3)
39 impulse(G*F_ramp,t)
40 title('Ramp Response') % need to set manually.
```

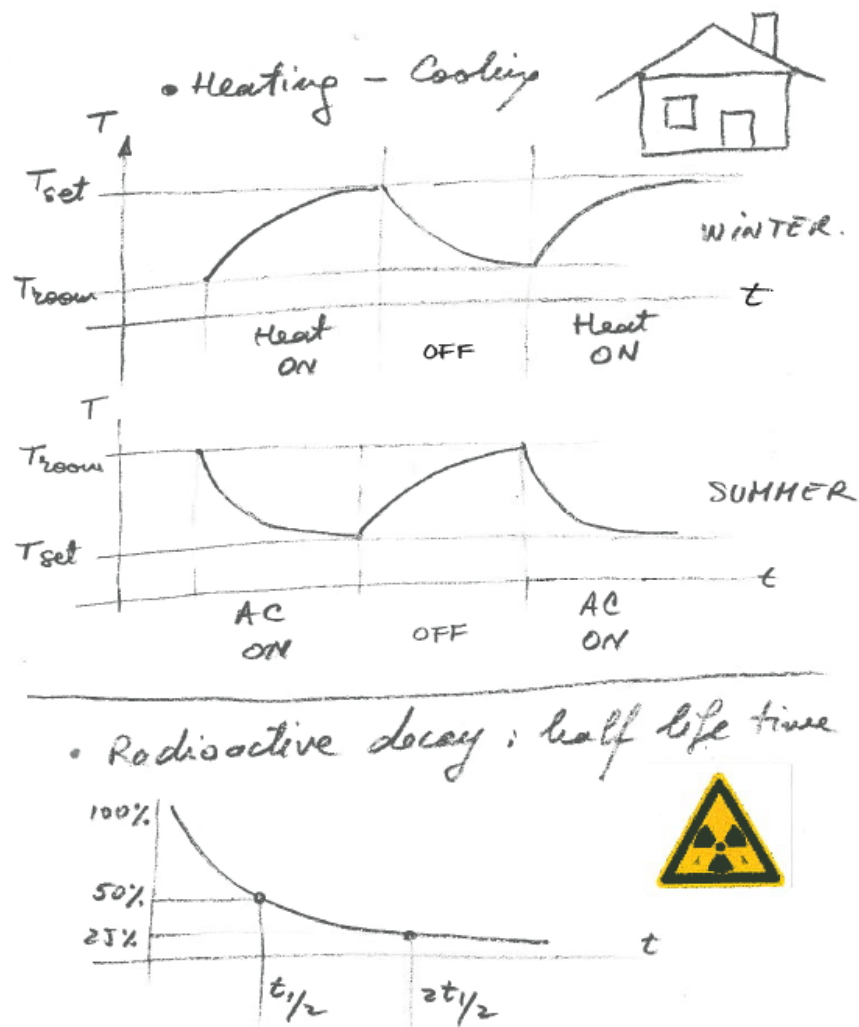


Figure 5.5: Examples of first order systems in the real world.

## 5.5 2<sup>nd</sup> Order System Time Response

The ordinary differential equation for the equation of motion of a 2<sup>nd</sup> order system can be expressed as

$$\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2x = \omega_n^2f(t) \quad (40)$$

with the initial conditions  $\ddot{x}(0) = 0$ , and  $\dot{x}(0) = 0$ . The Laplace transform gives us

$$\begin{aligned} x &\rightarrow X(s) \\ \dot{x} &\rightarrow sX(s) \\ \ddot{x} &\rightarrow s^2X(s) \\ f(t) &\rightarrow F(s) \end{aligned} \quad (41)$$

Taking the Laplace transform of the equation of motion yields

$$s^2X(s) + 2\zeta\omega_nsX(s) + \omega_n^2X(s) = \omega_n^2F(s) \quad (42)$$

Pulling  $X(s)$  out of the first equation results in

$$(s^2 + 2\zeta\omega_n s + \omega_n^2)X(s) = \omega_n^2 F(s) \quad (43)$$

next, we can solve for  $X(s)$

$$X(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} F(s) \quad (44)$$

As  $X(s) = G(s)F(s)$ , it we can show that

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (45)$$

### 5.5.1 Step Response for a 2<sup>nd</sup> Order System

$$X(s) = G(s)F(s) \quad (46)$$

$$\begin{aligned} &= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s} \\ &= \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)} \end{aligned} \quad (47)$$

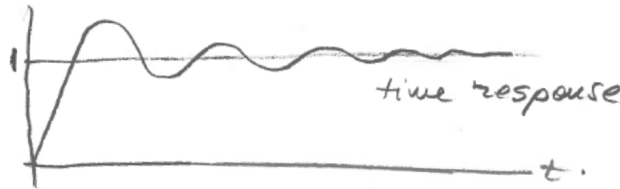
Therefore, solving for  $\mathcal{L}[X(s)]^{-1}$  yields

$$x(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t + \phi) \quad (48)$$

where

$$\begin{aligned} \phi &= \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta} \\ &= \sin^{-1} \sqrt{1-\zeta^2} \end{aligned} \quad (49)$$

or



Proof

$$7] x_c(t) = e^{-\zeta \omega_n t} (A \cos \omega_d t + B \sin \omega_d t) ; A, B \quad \text{ODE derivation}$$

$$= e^{-\zeta \omega_n t} C \sin(\omega_d t + \varphi)$$

$$\ddot{x} + 2\zeta \omega_n \dot{x} + \omega_n^2 x = \omega_n^2 f(t) ; \omega_n^2 f = \frac{f^*}{m} \rightarrow f = \frac{f^*}{k}$$

Step input,  $f(t) = 1(t)$

$$x_p(t) = D, \quad \dot{x}_p - \ddot{x}_p = 0$$

$$\omega_n^2 D = \omega_n^2 \rightarrow D = 1$$



$$x_p(t) = 1$$

$$x(t) = e^{-\zeta \omega_n t} C \sin(\omega_d t + \varphi) + 1$$

$$\dot{x}(t) = (-\alpha \sin \varphi + \omega_d \cos \varphi) C e^{-\alpha t}$$

$$x(0) = C \sin \varphi + 1 = 0$$

$$\dot{x}(0) = (-\alpha \sin \varphi + \omega_d \cos \varphi) = 0 \rightarrow \tan \varphi = \frac{\omega_d}{\alpha}$$

$$\tan \varphi = \frac{\omega_n \sqrt{1-\zeta^2}}{\zeta \omega_n} = \frac{\sqrt{1-\zeta^2}}{\zeta} \rightarrow \varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$$

$$\sin^2 \varphi = \frac{\tan^2 \varphi}{1 + \tan^2 \varphi} = \frac{1 - \zeta^2}{\zeta^2 + 1 - \zeta^2} = 1 - \zeta^2$$

$$\sin \varphi = \sqrt{1 - \zeta^2}$$

$$\varphi = \sin^{-1} \sqrt{1 - \zeta^2}$$

$$C = -\frac{1}{\sin \varphi} = -\frac{1}{\sqrt{1 - \zeta^2}}$$

$$x(t) = 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_d t + \varphi) \quad \text{unit response}$$

for  $f^* = k$

For  $f^* = F_0$

$$x(t) = \frac{F_0}{k} \left[ 1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_d t + \varphi) \right] = x_{st} \cdot x(t)$$

$$x_{st} = F_0/k \quad \text{static displacement}$$

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14) Step response  $f(t) = 1(t)$ ,  $F(s) = \frac{1}{s}$

$$X(s) = \frac{\omega_n^2}{(s+\alpha)^2 + \omega_d^2} \cdot \frac{1}{s} = \frac{A}{s} + \frac{Ds+E}{(s+\alpha)^2 + \omega_d^2}$$

ILT by  
partial  
fraction  
expansion

$$A(s+\alpha)^2 + A\omega_d^2 + Ds^2 + Es = \omega_n^2$$

$$A(s^2 + 2s\alpha + \alpha^2 + \omega_d^2) + Ds^2 + Es = 1$$

$$s^2: A + D = 0$$

$$s^1: 2\alpha A + E = 0$$

$$s^0: A(\alpha^2 + \omega_d^2) = \omega_n^2 \quad A(\cancel{s^2}\omega_n^2 + (1-\cancel{s^2})\omega_n^2) = \omega_n^2$$

$$A = 1$$

$$D = -A = -1$$

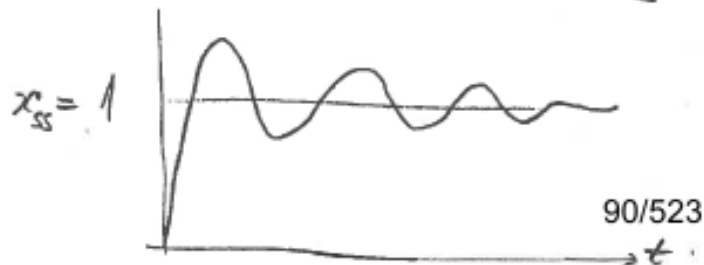
$$E = -2\alpha A = -2\alpha$$

$$X(s) = \left[ \frac{1}{s} - \frac{s+2\alpha}{(s+\alpha)^2 + \omega_d^2} \right] = \left[ \frac{1}{s} - \frac{\alpha + (s+\alpha)}{(s+\alpha)^2 + \omega_d^2} \right]$$

$$x(t) = 1 - e^{-\alpha t} \left[ \frac{\alpha}{\omega_d} \sin \omega_d t + \cos \omega_d t \right]$$

$$x(t) = 1 - e^{-\alpha t} \left( \frac{s}{\sqrt{1-\zeta^2}} \sin \omega_d t + \cos \omega_d t \right)$$

$$x(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta \omega_n t} \sin(\omega_d t + \varphi) \quad \varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$$



### 5.5.2 Impulse response of a 2<sup>nd</sup> Order System

$$X(s) = G(s)F(s) \quad (50)$$

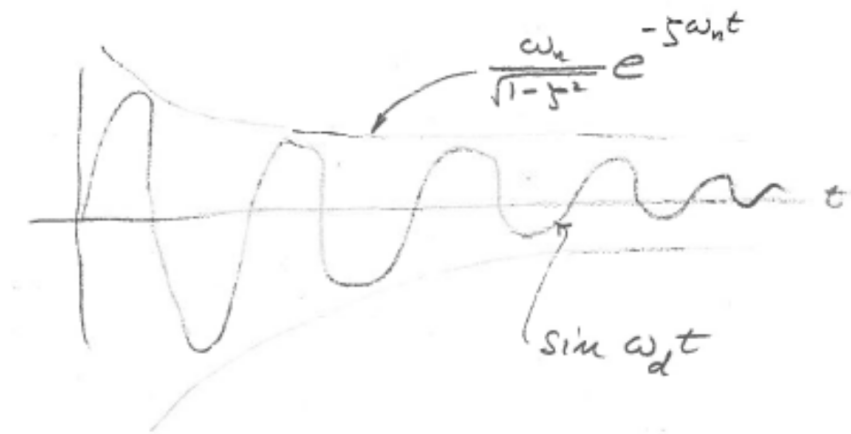
$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot 1$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (51)$$

Therefore, solving for  $\mathcal{L}[X(s)]^{-1}$  yields

$$x(t) = \frac{\omega_n}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \sin(\omega_d t) \quad (52)$$

or



Proof

13a) Long hand solution

ILT by partial fraction expansion

$$\frac{X}{\omega_n^2} = \frac{1}{s(s+p_1)(s+p_2)} = \frac{A}{s+p_1} + \frac{B}{s+p_2}$$

$$As + Ap_2 + Bs + Bp_1 = 1$$

$$s^1: A + B = 0 \rightarrow B = -A$$

$$s^0: Ap_2 + Bp_1 = 1 \rightarrow A = \frac{1}{p_2 - p_1}, B = \frac{1}{p_1 - p_2}$$

$$\frac{1}{(s+p_1)(s+p_2)} = \frac{1}{p_2 - p_1} \left( \frac{1}{s+p_1} - \frac{1}{s+p_2} \right)$$

$$x(t) = \frac{1}{p_2 - p_1} \left( e^{-p_1 t} - e^{-p_2 t} \right)$$

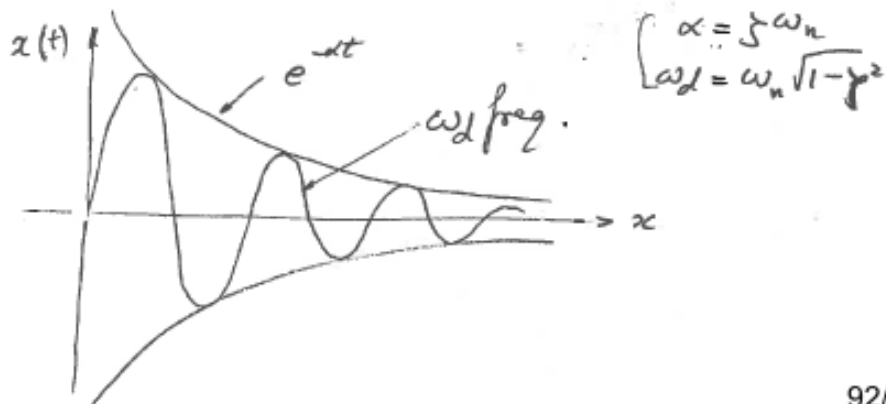
$$p_{1,2} = \zeta \omega_n \mp i \omega_n \sqrt{1 - \zeta^2} = \alpha \mp i \omega_d$$

$$p_2 - p_1 = i 2 \omega_d$$

$$\frac{x(t)}{\omega_n^2} = \frac{-1}{2i \omega_d} \left[ e^{-(\alpha - i \omega_d)t} - e^{-(\alpha + i \omega_d)t} \right]$$

$$= \frac{1}{2i \omega_d} e^{-\alpha t} \left[ e^{i \omega_d t} - e^{-i \omega_d t} \right]$$

$$= \frac{1}{\omega_d} e^{-\alpha t} \frac{e^{i \omega_d t} - e^{-i \omega_d t}}{2i} = \frac{e^{-\alpha t}}{\omega_d} \sin \omega_d t$$



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Alternative way

$$\frac{1}{(s+p_1)(s+p_2)} \xrightarrow{\mathcal{L}^{-1}} \frac{1}{p_2-p_1} (e^{-p_1 t} - e^{-p_2 t})$$

Another way      Residue Theorem

$$\frac{1}{(s+p_1)(s+p_2)} = \frac{a_1}{s+p_1} + \frac{a_2}{s+p_2}$$

$$a_1 = \left[ (s+p_1) \frac{1}{(s+p_1)(s+p_2)} \right]_{s=-p_1} = \frac{1}{-p_1+p_2}$$

$$a_2 = \left[ (s+p_2) \frac{1}{(s+p_1)(s+p_2)} \right]_{s=-p_2} = \frac{1}{p_1-p_2} = -a_1$$

### 5.5.3 Ramp response of a 2<sup>nd</sup> Order System

$$X(s) = G(s)F(s) \quad (53)$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \cdot \frac{1}{s^2}$$

$$= \frac{\omega_n^2}{s^2(s^2 + 2\zeta\omega_n s + \omega_n^2)} \quad (54)$$

Therefore, solving for  $\mathcal{L}[X(s)]^{-1}$  yields

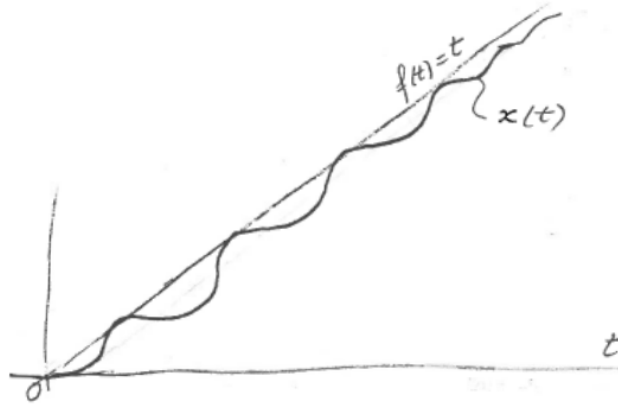
$$x(t) = t - \frac{2\zeta}{\omega_n} \left( 1 + \frac{1}{\sin \gamma_1} e^{-\zeta\omega_n t} \sin(\omega_d t - \gamma_1) \right) \quad (55)$$

where

$$\gamma_1 = \tan^{-1} \frac{2\zeta \sqrt{1-\zeta^2}}{1-2\zeta^2} \quad (56)$$

$$= \sin^{-1} 2\zeta \sqrt{1-\zeta^2} \quad (57)$$

or



Moreover,

$$\begin{aligned} \sin^2 \gamma_1 &= \frac{\tan^2 \gamma_1}{1 + \tan^2 \gamma_1} \\ &= \frac{4s^2(1-\zeta^2)}{(1-2\zeta^2) + 4\zeta^2(1-\zeta^2)} \\ &= \frac{4s^2(1-\zeta^2)}{1-4s^2+4s^4+4s^2-4s^4} \\ &= 4s^2(1-\zeta^2) \end{aligned} \quad (58)$$

and

$$\sin \gamma_1 = 2\zeta \sqrt{1-\zeta^2} \quad (59)$$

Proof

PROOF of 2<sup>nd</sup> order system Ramp response

ODE solution

$$x_p(t) = Dt + E$$

$$\dot{x}_p = D; \ddot{x}_p = 0$$

$$\ddot{x}_p + 2\zeta\omega_n \dot{x}_p + \omega_n^2 x_p = \omega_n^2 f(t)$$

$$2\zeta\omega_n D + \omega_n^2(Dt + E) = \omega_n^2 t$$

$$t^0: 2\zeta\omega_n D + \omega_n^2 E = 0 \rightarrow E = -\frac{2\zeta}{\omega_n} D$$

$$t^1: \omega_n^2 D = \omega_n^2 \rightarrow D = 1 \rightarrow E = -\frac{2\zeta}{\omega_n}$$

$$x_p(t) = t - \frac{2\zeta}{\omega_n}$$

$$x(t) = e^{-\zeta\omega_n t} C \sin(\omega_d t + \varphi_1) + t - \frac{2\zeta}{\omega_n}$$

$$\dot{x}(t) = (-\zeta\omega_n \sin\varphi_1 + \omega_d \cos\varphi_1) C e^{-\zeta\omega_n t} + 1$$

$$x(0) = 0: C \sin\varphi_1 - \frac{2\zeta}{\omega_n} = 0 \rightarrow C \omega_n \sin\varphi_1 = 2\zeta \quad (a)$$

$$\dot{x}(0) = 0: (-\zeta\omega_n \sin\varphi_1 + \omega_d \cos\varphi_1) C + 1 = 0$$

$$C(\omega_d \cos\varphi_1 - \zeta\omega_n \sin\varphi_1) = 1$$

$$\omega_d \cos\varphi_1 - \zeta\omega_n \sin\varphi_1 = \frac{1}{C}$$

$$(a): \quad 2\zeta^2 - C \omega_d \cos\varphi_1 = 1$$

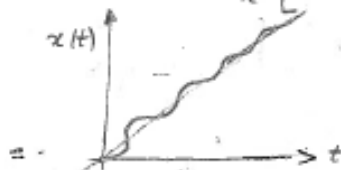
$$C \omega_n \sqrt{1-\zeta^2} \cos\varphi_1 = 2\zeta^2 - 1$$

$$C \omega_n \cos\varphi_1 = -\frac{1-2\zeta^2}{\sqrt{1-\zeta^2}} \quad (b)$$

$$\frac{(a)}{(b)} = \tan\varphi_1 = -\frac{2\zeta\sqrt{1-\zeta^2}}{1-2\zeta^2}$$

$$\varphi_1 = -\tan^{-1} \frac{2\zeta\sqrt{1-\zeta^2}}{1-2\zeta^2}, \quad C = \frac{2\zeta}{\omega_n} \cdot \frac{1}{\sin\varphi_1}$$

$$x(t) = t - \frac{2\zeta}{\omega_n} \left[ 1 - \frac{1}{\sin\varphi_1} e^{-\zeta\omega_n t} \sin(\omega_d t + \varphi_1) \right]$$



Note:  $\varphi_1$  is different from  $\varphi = \tan^{-1} \frac{\sqrt{1-\zeta^2}}{\zeta}$  of step response

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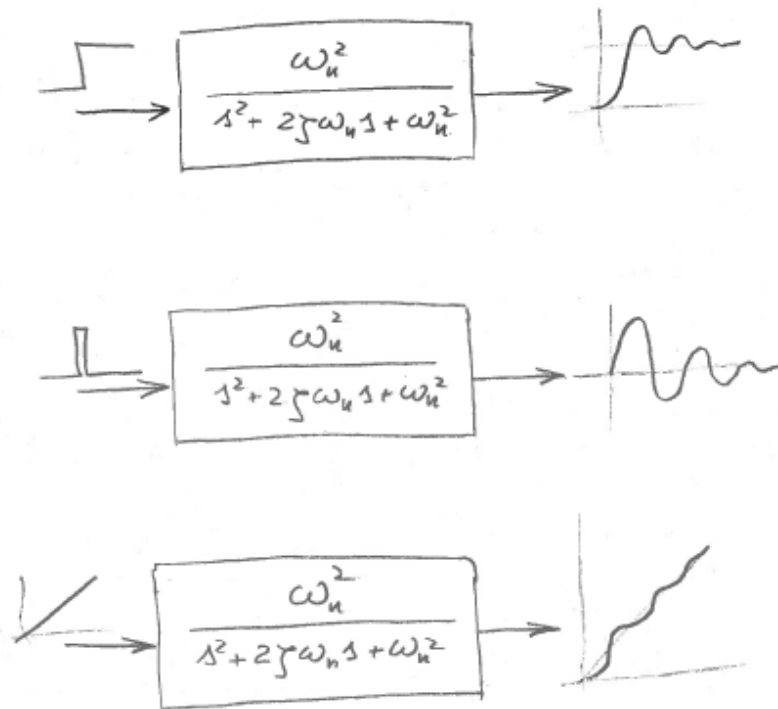
**5.5.4 Summary of the Second Order System Responses**

Figure 5.6: A summary of the second order system responses.

Listing 8: MATLAB code for time series responses of 2<sup>nd</sup> order system.

---

```

1  %{
2  This program studies time response of 2nd order systems
3  %}
4
5  clc
6  clear
7  close all
8
9  format compact
10
11 %% Given data
12 fn = 5; % time response for the 1D system
13 wn = 2*pi()*fn
14 z = 0.035
15
16 %% time range setup
17 T_max = 10; % run the test to 10 seconds
18 dt = T_max*1e-4; % find the delta-t value
19 t = 0:dt:T_max; % build the time vector
20
21 %% Define system
22 B = [wn^2];
23 A = [1 2*z*wn wn^2];
24 G = tf(B,A);
25
26 %% create the figure enviornment
27 figure(1)
28 xlim([0 1])
29
30 %% step response
31 subplot(3,1,1)
32 hold on
33 step(G,t)
34 ylim([0 2])
35
36 %% Impulse response
37 subplot(3,1,2)
38 impulse(G,t)
39 ylim([-35 35])
40
41 %% Ramp response
42 F_ramp = tf([1],[1 0 0])
43 subplot(3,1,3)
44 impulse(G*F_ramp,t)
45 xlim([0 1])
46 ylim([0 1])
47 title('Ramp Response') % need to set manually.

```

---